

Lab 10 Report

Ryan Wilson

Due: 04/21/2026 @ 11:00 AM

1 Complexity

The time complexity for the Naive-Gauss elimination in terms of the number of equations (n) is $\mathcal{O}(n^3)$. This is because there are three nested loops, meaning there are three operations successively using n , resulting in the complexity being $\mathcal{O}(n^3)$.

2 Pseudo-code

Algorithm 1 Naive Gaussian Elimination

```
1: for  $j = 0$  to  $n$  do
2:   for  $i = 0$  to  $n$  do
3:     if  $i \neq j$  then
4:        $factor = \frac{A_{ij}}{A_{jj}}$ 
5:       for  $k = 0$  to  $n + 1$  do
6:          $A_{ik} = A_{ik} - factor \cdot A_{jk}$ 
7:       end for
8:     end if
9:   end for
10: end for
```

3 Sources

<https://www.geeksforgeeks.org/computer-science-fundamentals/finding-inverse-of-a-matrix-using-gauss-jordan-method/>
<https://www.geeksforgeeks.org/c/how-to-use-make-utility-to-build-c-projects/>
<https://www.geeksforgeeks.org/python/gauss-seidel-method/>
<https://www.geeksforgeeks.org/c/c-program-print-contents-file/>